“Can you look into the sales data?” is one of the most common requests an analyst receives, and it is not yet a question. Answering it exactly as written is how a week disappears.

CHAPTER 3 — THE DATA ANALYSIS PROCESS

Defining the Problem and Asking Questions

Lesson 1 of 6

BY THE END

You can turn a vague request into a specific analytical question.

PLATE I — THE DATA ANALYSIS PROCESS



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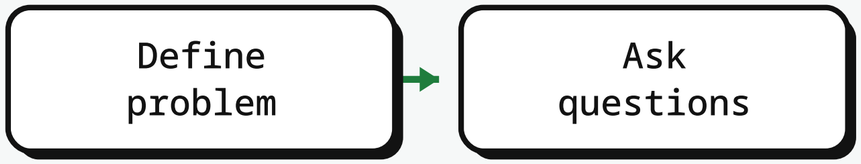


PLATE I — THE DATA ANALYSIS PROCESS



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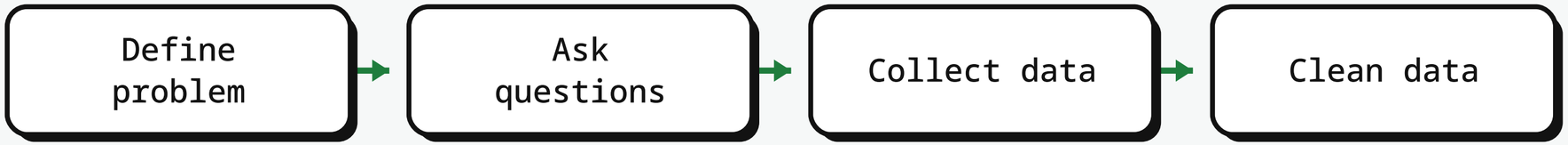


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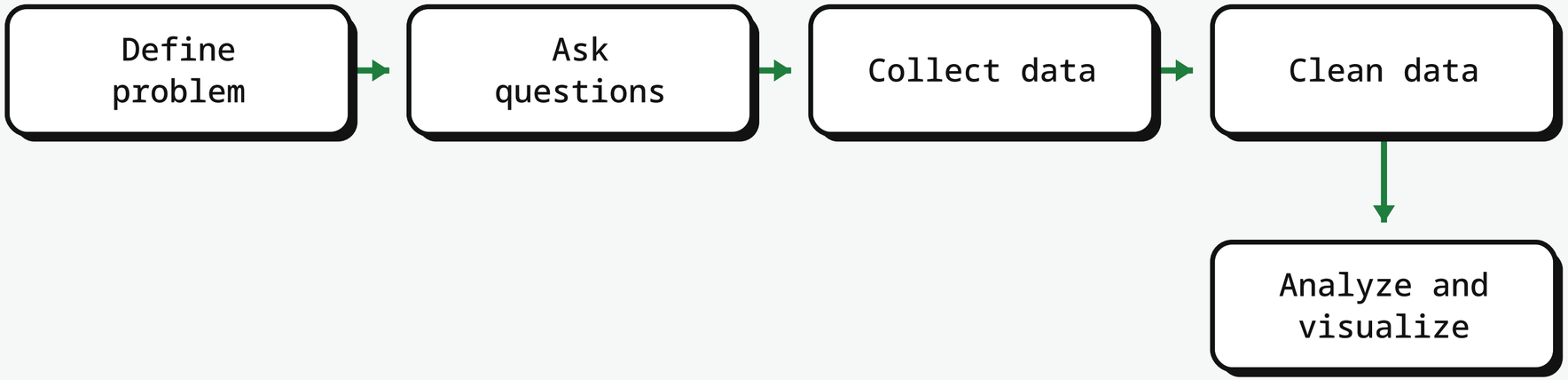


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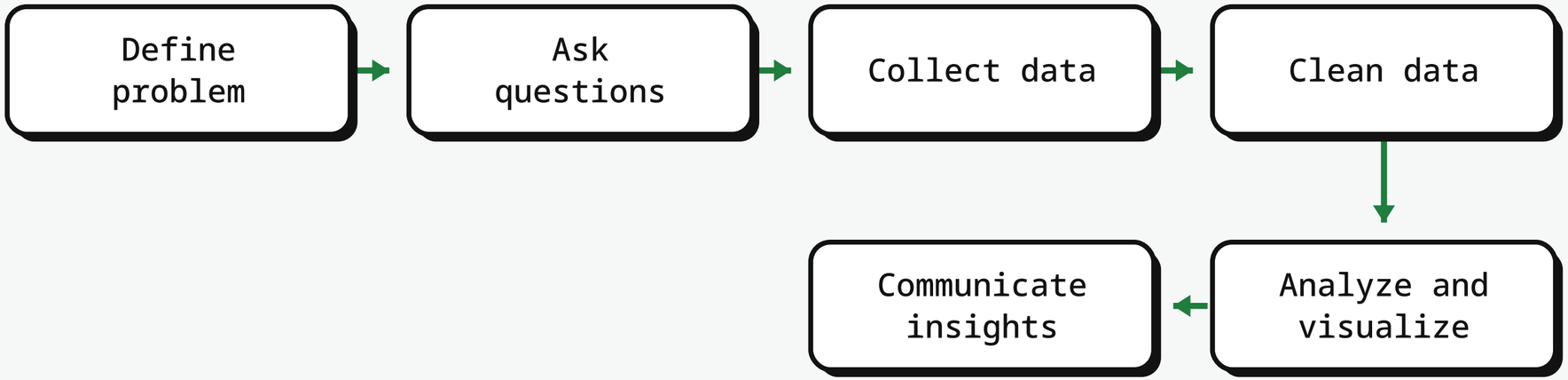
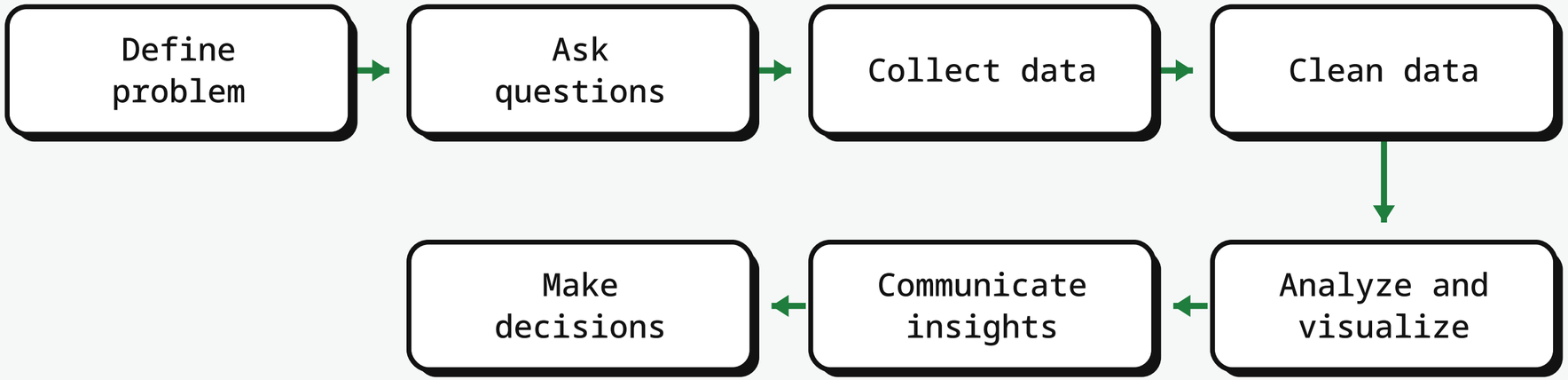


PLATE I — THE DATA ANALYSIS PROCESS



every stage depends on the question you asked first

Start with the business problem

A common mistake is jumping straight into data without knowing what you are trying to solve. Always begin by asking:

- What problem are we trying to solve?
- Who cares about the answer?
- What decision will this analysis support?

Start with the business problem

EXAMPLE

- Weak question: “Let’s analyze the sales data.”
- Strong question: “Why did sales drop in the North region last quarter?”

FOR A LOCAL SHOP, THE SAME IDEA APPLIES

- Weak question: “Let’s look at the sales spreadsheet.”
- Strong question: “Why did rice sales drop in the last two weeks compared to the month before?”

The stronger questions give analysis a clear direction.

Know your stakeholders

A **stakeholder** is anyone who needs the results of your analysis.

EXAMPLES

- A manager deciding on marketing spend
- A product team improving an app feature
- A finance team reviewing budget performance

UNDERSTANDING STAKEHOLDERS HELPS YOU

- Ask the right questions
- Choose relevant metrics
- Present results in a useful way

How to ask good analytical questions

GOOD QUESTIONS ARE

- `Specific` — focused on one issue
- `Measurable` — answerable with data
- `Relevant` — connected to a real decision
- `Clear` — easy for others to understand

EXAMPLES

- `Which product category had the highest return rate last month?`
- `Are new customers more likely to cancel within 30 days?`
- `Which marketing channel brings the most qualified leads?`

4 — WORKED EXAMPLE

An online course platform notices fewer enrollments.

STAKEHOLDERS

marketing team and product manager

PROBLEM

enrollment decline

QUESTIONS

- When did enrollments start dropping?
- Did the drop affect all courses or only some?
- Did traffic decrease, or did fewer visitors convert?

KEY TERMS

TERM	DEFINITION
Business problem	The issue or decision the analysis is meant to support
Stakeholder	A person or group affected by, or interested in, the results of the analysis
Analytical question	A clear question that data can help answer